

# Appunti Network Analysis

## Formule Network Analysis – Cheat Sheet

| Concept                    | Formula                                                                          |
|----------------------------|----------------------------------------------------------------------------------|
| Average degree             | $\frac{2E}{N}$                                                                   |
| Maximum edges (undirected) | $\frac{N(N-1)}{2}$                                                               |
| Maximum edges (directed)   | $N(N-1)$                                                                         |
| Density (undirected)       | $\frac{2E}{N(N-1)}$                                                              |
| Density (directed)         | $\frac{E}{N(N-1)}$                                                               |
| Degree centrality          | $\frac{\text{degree of node}}{N-1}$                                              |
| Closeness centrality       | $\frac{N-1}{\text{total distance from node to all other nodes}}$                 |
| Betweenness centrality     | $\frac{\text{shortest paths passing through node}}{\text{total shortest paths}}$ |
| Normalized betweenness     | $\frac{\text{betweenness}}{\frac{(N-1)(N-2)}{2}}$                                |
| Local clustering           | $\frac{\text{edges between neighbors}}{\text{possible edges between neighbors}}$ |
| Global clustering          | $\frac{3 \times \text{triangles}}{\text{connected triplets}}$                    |
| Eccentricity               | largest distance from node to any other node                                     |
| Diameter                   | largest eccentricity in the graph                                                |
| Degree distribution        | $\frac{\text{nodes with degree } k}{N}$                                          |
| ER expected degree         | $p(N-1)$                                                                         |
| ER expected edges          | $p \frac{N(N-1)}{2}$                                                             |
| ER expected density        | $p$                                                                              |
| Giant component fraction   | $\frac{\text{size of largest component}}{N}$                                     |

Table 1: Network Analysis Formula Sheet